FINA 60201A – Fixed-income Securities

Second Home Assignment

Winter 2026

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PART ONE

The objective is to compute the term structure of credit spreads for the company Johnson & Johnson (GVKEY 006266, PERMCO 21018, PERMNO 22111).

1] From Compustat Fundamentals Quarterly, download the series of long-term debt $(DLTTQ_t)$ and debt in current liabilities $(DLCQ_t)$ from 2024Q4 to 2025Q4. Use a linear interpolation to construct the daily series for $L_t \coloneqq DLCQ_t + 0.5DLTTQ_t$.

2] From Crisp, download the daily time series of market capitalization.

3] Apply the iterative approach to estimate the asset value and volatility by the end of 2025. Use the daily one-year zero-coupon yield from the St-Louis Fed as the risk-free rate. Make sure the trading days coincide between the market capitalization and the risk-free rate.

4] Assuming that L_t and $DLTTQ_t$ are the default points at the one- and twenty-year horizons, respectively, use a linear inter- and extrapolation to construct the default frontier for all maturities between 0 and 30 years.

5] On the first trading day of 2026, apply the Merton model to compute the term structure of credit spreads (up to 30 years). Use the zero-coupon yields from the St-Louis Fed as the risk-free rates.

PART TWO

The St-Louis Fed website publishes monthly series of zero-coupon yields for "high quality market" corporate bonds, defined as a set of corporate bonds rated AAA, AA or A. We will use these data as a proxy for the term structure of AA yields. By the end of February 2026, the following yields for 11 maturities were reported:

Maturity	0.5	1	2	3	4	5	7	10	15	20	30
Yield (%)	3.76	3.79	3.86	3.96	4.08	4.20	4.52	4.83	5.42	5.69	6.01

1] Download from the St-Louis Fed website the Nelson-Siegel-Svensson function estimated on February 27, 2026. Use these risk-free zero-coupon yields $zcy(t)$ to construct the credit spreads for the 11 maturities. Plot the three term structures: risk-free, AA yields, and credit spreads.

2] Consider the Duffee (1999) model where the EMM dynamics of the risk-free short rate r_t and of the credit spread λ_t are given by:

$$dr_t = \kappa_r(\theta_r - r_t)dt + \sigma_r\sqrt{r_t}dW_t,$$

$$ds_t = \kappa_s(\theta_s - s_t)dt + \sigma_s\sqrt{s_t}dZ_t,$$

$$dW_t dZ_t = 0,$$

$$\lambda_t = \alpha + \beta r_t + s_t.$$

Write down the expression for the yield on the defaultable zero-coupon bond, denoted by $Y(t)$. Hint: If r_t is a square-root process, then $(1+\beta)r_t$ is still a square-root process with same speed of mean reversion, long-term mean multiplied by $(1+\beta)$ and volatility multiplied by $\sqrt{1+\beta}$.

3] Calibrate the Duffee (1999) model in two steps.

- Fit the theoretical risk-free zero-coupon yields with the 11 zero-coupon yields $zcy(t)$ to obtain the parameters r_0, κ_r, θ_r and σ_r.
- Fit the theoretical defaultable zero-coupon yields $Y(t)$ with the 11 observed AA yields to obtain the parameters s_0, κ_s, θ_s, σ_s, α and β.

Check the quality of the fit by plotting on the same graph the AA yields and the continuous function $Y(t)$ for maturities ranging from 0 to 30 years.

4] Use the calibrated Duffee (1999) model to evaluate the following AA corporate bond issue:

- Nominal: $1,000,000.
- Maturity: 5 years.
- Coupon rate: 4% paid semi-annually.
- Callable at par at anytime.

Use the explicit finite difference scheme in two dimensions (for r_t and s_t).

5] Compute the yield-to-maturity on the callable bond. Compare it with the yield-to-maturity on the similar non-callable bond issue (use the closed-form solution).